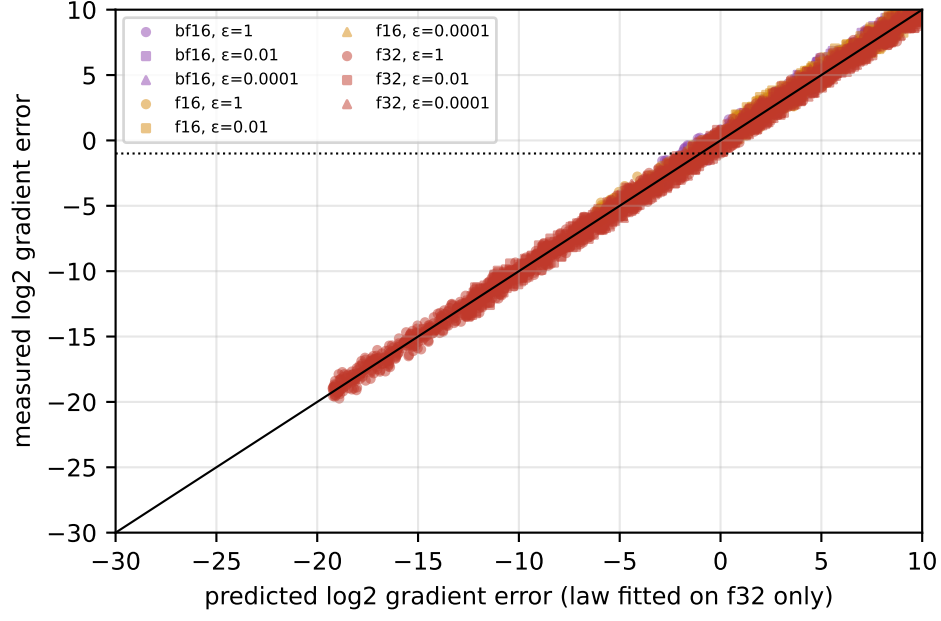
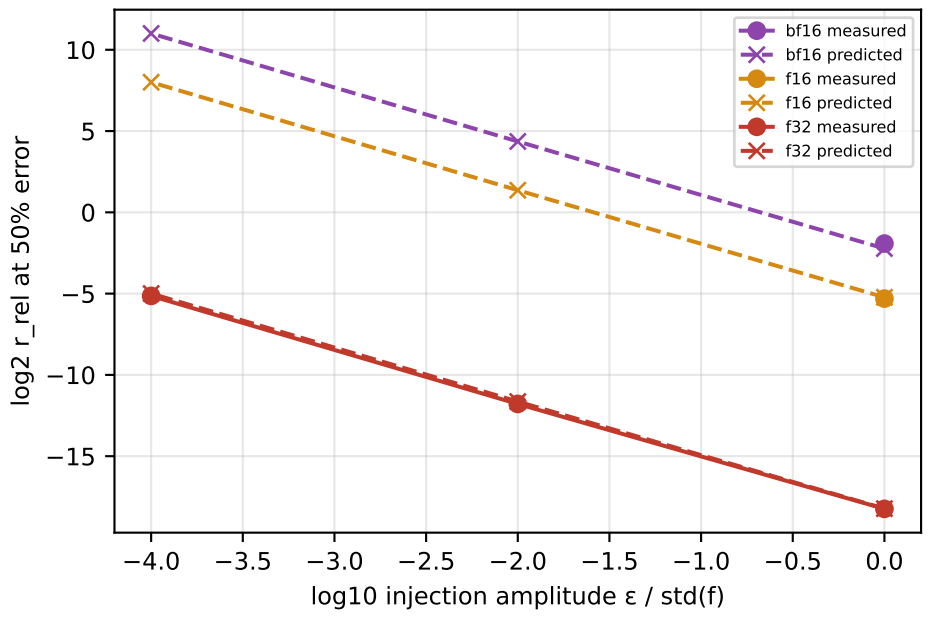

D1: one law across precisions
bf16, ε=1
bf16, ε=0.01
bf16, ε=0.0001
f16, ε=1
f16, ε=0.01
f16, ε=0.0001
f32, ε=1
f32, ε=0.01
f32, ε=0.0001
measured log2 gradient error
predicted log2 gradient error (law fitted on f32 only)
10
5
0
−5
−10
−15
−20
−25
−30


D1: resolvability threshold
bf16 measured
bf16 predicted
f16 measured
f16 predicted
f32 measured
f32 predicted
log2 r_rel at 50% error
log10 injection amplitude ε / std(f)
10
5
0
−5
−10
−15
−4.0
−3.5
−3.0
−2.5
−2.0
−1.5
−1.0
−0.5
0.0